Collective Convergence on Institutional Structures: Progress Report

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Abstract

This progress report documents our work investigating how agents with uncertain beliefs about collective structure converge on shared institutional understanding through Bayesian inference. We report on our implementation of baseline coordination models and outline remaining work to incorporate asymmetry and inverse planning approaches.

1 Progress to Date

1.1 Learning MeMo Framework

We have completed hands-on tutorials with the MeMo probabilistic programming framework, which enables recursive Bayesian modeling of multi-agent systems. This provides the computational foundation for modeling agents who reason about each other's beliefs about institutional roles.

1.2 Baseline Symmetrical Model

We implemented a baseline two-agent coordination model in the MeMo framework exploring different levels of strategic reasoning:

- **Random baseline**: Agents choose actions uniformly at random
- **Naive game**: Agents have random role assignments and act according to roles
- **Hierarchical game**: Agents have asymmetric role preferences (leader vs. follower)
- **One-sided rational**: One agent models the other's behavior
- **Mutual rational**: Both agents model each other as naive
- **Recursive reasoning**: Agents model each other at varying depths of mutual reasoning

Preliminary finding: All models converge to similar collective payoffs (1.0) because the current setup is perfectly symmetrical. Without asymmetry-breaking mechanisms, agents cannot stably differentiate into complementary roles.

2 Remaining Work

2.1 Phase 1: Breaking Symmetry (Deadline: Oct 24)

Joseph: Implement asymmetric role priors where agents have slight biases toward different roles (e.g., Alice 55% leader preference, Bob 55% follower preference). Test whether recursive reasoning amplifies these small differences into stable role differentiation.

Emiel: Add observation-based learning where agents update role beliefs based on interaction outcomes over multiple rounds. Measure convergence speed to stable role assignments.

2.2 Phase 2: Inverse Planning Implementation (Deadline: Nov 7)

Joseph: Reformulate the current game-theoretic approach using inverse planning where agents infer others' roles from observed actions rather than best-responding to predicted actions. Compare convergence properties of both approaches.

Emiel: Extend the model to N agents (3-5) to test whether hierarchical structures emerge in larger groups and how inference scales with group size.

2.3 Phase 3: Theoretical Integration (Deadline: Nov 21)

Joseph: Revisit (Jara-Ettinger and Dunham 2024) and formalize connections between our computational model and the institutional stance framework, specifically how role-based normative expectations constrain the inference process.

Emiel: Integrate insights from (Davis, Dunham, and Jara-Ettinger 2022) on social structure inference, particularly their "intuitive sociologies" likelihood function P(D|S), to ground our model in established theory.

2.4 Phase 4: Notebook Finalization (Deadline: Dec 1)

Both team members: Complete interactive Jupyter notebook with visualizations and analysis (see outline below).

3 Notebook Outline

3.1 Introduction and Motivation

Overview of institutional convergence problem with simple illustrative example. Contrast with traditional multi-agent coordination approaches.

3.2 MeMo Framework Basics

Brief tutorial on key MeMo primitives (chooses, thinks, E, wpp) with minimal working examples to make notebook self-contained.

3.3 Baseline Models: Symmetric Case

Progressively build from random agents through naive, one-sided rational, mutual rational, to recursive reasoning models at varying depths. Show that symmetry prevents stable role differentiation. Include visualization of action probabilities and expected payoffs across all model variants.

3.4 Breaking Symmetry: Asymmetric Priors

Introduce slight role biases and demonstrate how recursive reasoning amplifies differences. Visualize belief trajectories over reasoning depth.

3.5 Breaking Symmetry: Observational Learning

Multi-round interaction where agents update beliefs based on observed outcomes. Plot convergence dynamics and final role assignments across different initial conditions.

3.6 Game-Theoretic vs. Inverse Planning

Compare the two approaches side-by-side on the same coordination problems. Analyze computational costs and convergence properties.

3.6.1 Game-Theoretic Approach

Best-response dynamics and equilibrium analysis.

3.6.2 Inverse Planning Approach

Bayesian role inference from observed actions.

3.6.3 Comparative Analysis

Which approach converges faster? More reliably? Under what conditions does each excel?

3.7 Scaling to N Agents

Extend models to 3-5 agents. Visualize emergent hierarchical structures using network diagrams. Analyze how inference complexity scales.

3.8 Theoretical Connections

Discuss connections to institutional stance ((Jara-Ettinger and Dunham 2024)) and social structure inference ((Davis, Dunham, and Jara-Ettinger 2022)). Interpret computational results through these theoretical lenses.

3.9 Discussion and Future Directions

Summarize findings, limitations, and promising directions for future work.

Bibliography

[1] J. Jara-Ettinger and Y. Dunham, “The institutional stance,” *Under review*, 2024.

[2] I. Davis, Y. Dunham, and J. Jara-Ettinger, “Inferring the internal structure of social collectives,” 2022.